Task 1: Data Audit Report

Employee attrition dataset (CortexHR / Aura Workforce) | Prepared by Katarukonda Ashok (GM00154) | October 2026

1. In short

I checked the attrition table before building any model. It is **usable but not ready**: it needs a small clean-up, five columns must be removed because they reveal the answer, and the model must be judged with the right metrics because only about 1 in 7 employees leaves. Every problem below has a written decision, and those decisions become the rules of the Task 2 pipeline.

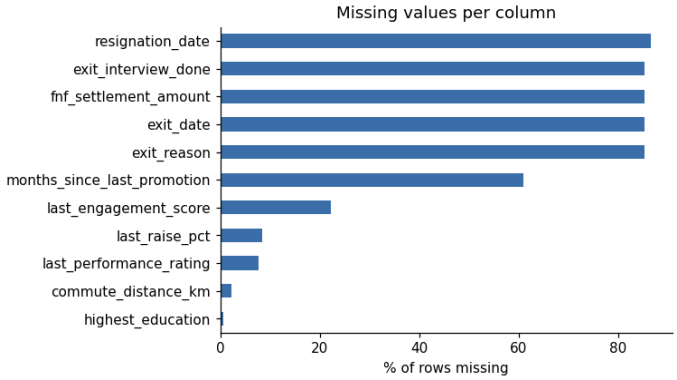
What I checked	What I found
Size	2,389 rows and 42 columns, one row per employee (snapshot 30 Sep 2025 plus a historical extension from 2021-2025). 6 rows were exact copies, leaving **2,383 employees**.
Missing values	11 columns have gaps. Most are not data loss: they mean "not applicable" (never promoted, no raise yet, not reviewed yet).
Outliers	Two methods (IQR and z-score) disagree a lot. Only a handful of rows are real errors; most flagged rows are genuine senior staff or overtime workers.
Errors	3 salaries 10x too high, 4 negative tenures, 2 attendance rates above 100%, 13 inconsistent city spellings, 10 negative experience values, 90 rows where tenure is longer than total experience.
Imbalance	Only **14.6%** left (349 people). A model that always says "stays" would be 85.4% accurate and useless.
Leakage	**6 columns** reveal the answer (exit date, resignation date, exit reason, exit interview, settlement, and the voluntary-exit flag).

2. Missing values

I first asked *why* each value is missing, because the right fix depends on the reason.

Column	Missing	Why it is missing	Decision
months_since_last_promotion	61.0%	Never promoted (explains 100% of gaps)	Fill + add "never promoted" flag. Keep the column
last_engagement_score	22.2%	Did not take the survey; these people leave slightly more (16.7% vs 14.1%)	Add "missing" flag, then median fill
last_raise_pct	8.4%	No raise received yet (100% of gaps)	Fill 0 + flag
last_performance_rating	7.6%	Recent joiners not yet reviewed (average tenure about 5 months)	Flag, then median fill
commute_distance_km	2.3%	Random gaps	Median fill by city
highest_education	0.6%	Random gaps	Category "Unknown"
5 exit columns	85-87%	Blank by design for people who stayed	Drop (leakage, section 5)

If a column were 40% missing, what would I do? I would not drop it straight away. I would find out why it is blank. If the blank itself means something (like "never promoted"), I fill it sensibly and add a "was missing" flag so the signal is kept. I would drop it only if it is mostly empty and unrelated to leaving, or if it leaks the answer.

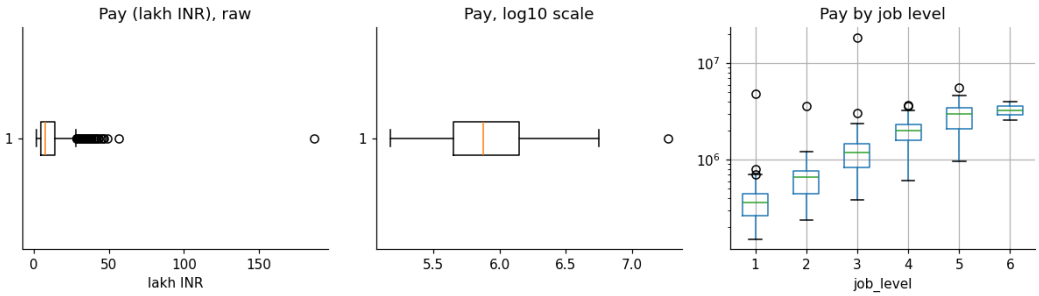


3. Outliers: IQR versus z-score

An outlier is an unusual value. It is only a problem if it is an *error*. I compared two methods on the key numeric columns.

Column	IQR	Z-score	What it really is
annual_fixed_pay	93	26	Right-skewed. Almost all are senior staff. On a log scale or within a job level only 1-9 remain. **3 are real typos**
avg_overtime_hours_per_month_3m	445	61	81% have zero overtime, so IQR flags every overtime worker. False alarm
avg_hours_worked_3m	48	15	One extreme record (EMP-0313: 1.8 hours, 20% attendance). Flag for HR
tenure_months	73	45	Long tenures up to 164 months are genuine. 4 negative values are errors

Lesson: IQR assumes a roughly symmetric spread, so it over-flags skewed data. Z-score is less sensitive but is pulled by the outliers themselves. Neither method says "this is an error"; a business rule does. I therefore **keep genuine high values** and use *log(pay)* and *comp_ratio*, and I **fix only values that are impossible**.



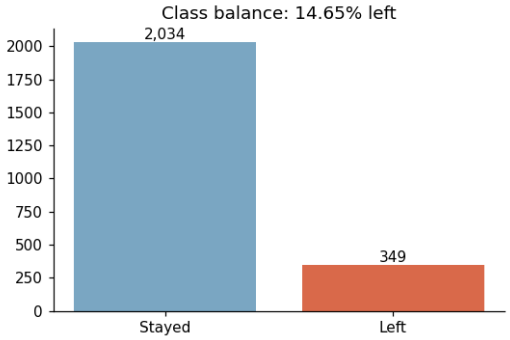
4. Errors found and how I will fix them

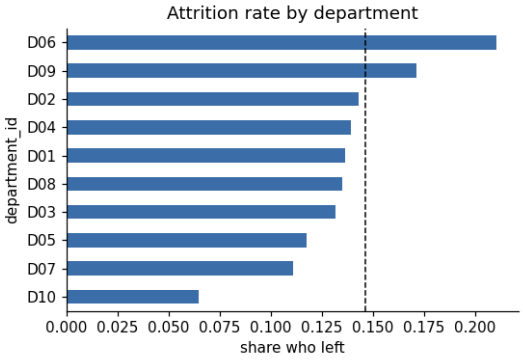
Problem	Rows	Fix
Salary about 10x too high (EMP-0034, EMP-0433, EMP-0601); no longer matches comp_ratio	3	Divide by 10 and confirm with the Python team
Negative tenure (join date after observation date)	4	Set to 0 and flag for source check
Attendance rate 1.12 (above 100%)	2	Clip to 1.0
City spelled differently (BENGALURU, Bangalore, Madras, delhi ncr...)	13	Standardise to one spelling
wfh_share outside 0-1 (small noise on Onsite and Remote staff)	213	Clip to the range 0-1
Negative total experience (historical rows)	10	Clip at 0
Tenure more than 6 months longer than total experience	90	Raise experience to match tenure
Exact duplicate employees	6	Drop the copies

5. Class imbalance

Of 2,383 employees, **349 left (14.6%)** and 2,034 stayed, about **5.8 stayers for every leaver**. Attrition also varies by department: D06 is highest at 21% and D10 lowest at 6.5%.

- Accuracy is misleading. "Nobody leaves" is 85.4% accurate yet finds zero leavers.
- I will judge models with **recall, precision, F1, ROC-AUC** (and PR-AUC), using stratified splits and class weights.
- For HR, **recall** matters most: missing a leaver costs a resignation, while a false alarm only costs a conversation. Precision still has to stay reasonable so HR keeps trusting the list.



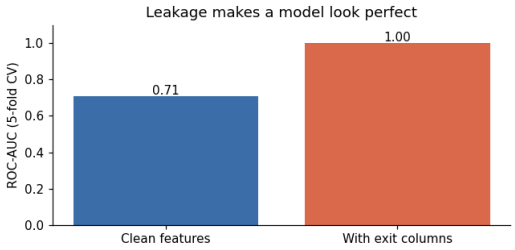


6. Data leakage: what must not reach the model

Leakage means a column contains information that would not exist at the time we need to predict. Like studying with the answer key, the model looks perfect but fails in real life.

Column	Why it leaks	Decision
exit_date, resignation_date, exit_reason, exit_interview_done, fnf_settlement_amount	Filled for 92-100% of leavers and 0% of stayers. Created at exit	Drop
voluntary_left_within_12m	A subset of the target itself	Drop
row_id, employee_id, source	Identifiers and how the row was created, not real signal	Exclude
dept_attrition_rate_prior_12m, comp_ratio, attendance windows	Measured on or before the snapshot date	Safe to use

Proof: a simple model on clean features scores ROC-AUC **0.71**. Adding just two exit columns lifts it to **1.00**. A perfect score is a warning sign, not a success. The strongest single clean feature (engagement score) reaches only 0.61, so a realistic final model should land near 0.70-0.75.



7. Next steps (Task 2 pipeline rules)

- Fit imputers, scalers and encoders on **training data only**, never on the whole dataset.
- Dates span 2021-2025, so test a **time-based split** as well as stratified random CV.
- Do not use gender or age (separate sheet) without a fairness review.
- Open items for the team: confirm the 3 salary corrections, the 4 negative tenures and EMP-0313 with the Python team.